

Panagiotis Chatzigiannis

Resume

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📄 <https://panoschtz.github.io>

Summary

Research Scientist specializing in cryptography, blockchains, privacy-preserving systems, decentralized payments, auditability, accountability, and Web3 recovery. Experience spans industrial research, academic research, and security-focused engineering in military and applied cryptography settings.

Experience

June 2022 – **Research Scientist**, *Visa Research*.

- Present
 - Research on cryptography, privacy-preserving financial technologies, blockchain security, and decentralized systems.
 - Contributed to projects on Web3 recovery, payment channels, proof-of-stake security, and privacy-enhancing analytics.
 - Published research across applied cryptography, blockchain, and financial privacy topics.

May 2021 – **Research Intern**, *Facebook (Diem)*.

- Sep 2021
 - Studied blockchain light clients and systematized techniques for efficient verification.
 - Investigated compression methods for cryptographic primitives.

May 2020 – **Research Intern / Part-Time Student Researcher**, *Facebook (Diem)*.

- Dec 2020
 - Worked on Proof of Assets for blockchain compliance use cases.
 - Developed methods for compressing discrete-log lookup tables with low overhead.

2001 – 2017 **Lieutenant Commander**, *Greek Navy*.

- Chief engineer and electronics officer on submarines and frigates.
- Served in IT leadership, cyber-defense, and incident response roles.
- Supported NATO maritime operations and technical training environments.

Education

2017 – 2022 **Ph.D. in Computer Science**, *George Mason University*, Fairfax, VA, GPA: 3.92.

Advisor: Prof. Foteini Baldimtsi

2010 – 2012 **M.Sc. in Computer Science**, *Naval Postgraduate School*, Monterey, CA, GPA: 3.82.

Thesis: *Connecting Land-based Networks to Ships*

1997 – 2001 **B.Sc. in Engineering**, *Hellenic Naval Academy*, Piraeus, Greece.

Selected Publications

2025 SoK: Fully-homomorphic Encryption in Smart Contracts

- 2025 A Composability Analysis Framework for Web3 Wallet Recovery Mechanisms
- 2024 A Plug-and-Play Long-Range Defense System for Proof-of-Stake Blockchains
- 2024 Truncator: Time-space Tradeoff of Cryptographic Primitives
- 2023 SoK: Web3 Recovery Mechanisms
- 2023 Privacy-Enhancing Technologies for Financial Data Sharing
- 2022 SoK: Blockchain Light Clients
- 2022 gOTzilla: Efficient Disjunctive Zero-Knowledge Proofs with Application to Proofs of Assets
- 2021 MiniLedger: Compact-sized Anonymous and Auditable Distributed Payments
- 2021 SoK: Auditability and Accountability in Distributed Payment Systems

Service

- Program Committees PETS 2023, 2025, 2026; Financial Cryptography 2023–2026; ICBC 2025; CCS 2026
- Reviewing NDSS, ACM CCS, Eurocrypt, Financial Cryptography, AFT, and related venues
- Academic Roles Research Assistant (2018–present); Teaching Assistant for Network Security, Computer Architecture, and Introduction to Computer Programming

Awards

- 2023 PETs Challenge 2022 – Special Recognition Prize
- 2016 National Bank of Greece Fintech Hackathon
- 2016 Panoptis National Cyberdefense Exercise – Navy commendation

Technical Skills

- Languages Python, Rust, C, C++, Java, Go, Solidity, Arduino
- Tools Wireshark, Oracle, MSSQL, MySQL, Cisco IOS
- Areas Cryptography, blockchain security, privacy-preserving systems, network security, decentralized payments

Languages

Greek (Native), English, German